

Kadirov Quant Lab

The Autonomous Multi-Agent Hedge Fund System

Executive Summary

Traditional quantitative funds rely on static trading models that degrade as market regimes shift. **Kadirov Quant Lab** solves this alpha decay with a hierarchical, multi-agent AI framework that transforms quantitative trading into a continuous, self-evolving loop. By unifying research, backtesting, code generation, and live execution into a single agentic workflow, the Lab ensures trading strategies adapt to real-time market realities without losing mathematical rigor or risk compliance.

The Problem

- **Alpha Decay:** Profitable trading models inevitably lose efficacy as market conditions change, requiring slow, manual human intervention to recalibrate.
- **Siloed Workflows:** The traditional quant pipeline—moving from thesis to historical backtest, code deployment, and risk monitoring—is fragmented, prone to human error, and slow to deploy.
- **Qualitative Blindspots:** Purely data-driven AI systems often fail to account for unprecedented macro events, geopolitical shifts, or nuanced human insights.

Our Solution: The Agentic Workflow

Kadirov Quant Lab replaces static models with an autonomous, continuous improvement cycle driven by specialized, hierarchical AI agents:

- **Autonomous Auditing & Self-Correction:** Specialized agents constantly audit live performance against deep historical data. When a strategy deviates from expected risk-reward profiles, the system autonomously rewrites and self-corrects the underlying trading logic in real-time.
- **Unified Pipeline:** Research, backtesting, code generation, and live execution are compressed into a single, seamless agentic loop, drastically reducing time-to-market for new alpha generation.
- **Human-in-the-Loop (HITL) Co-Evolution:** A proprietary translation layer allows fund managers to input qualitative insights (e.g., geopolitical risk, macro-economic pivots). The system instantly converts these insights into rigorous mathematical constraints, ensuring human and AI intelligence co-evolve.

Market Opportunity & Edge

The systematic hedge fund industry is crowded, but heavily reliant on human-driven coding cycles. Kadirov Quant Lab sits at the intersection of Generative AI and Advanced Quantitative Finance:

Feature	Traditional Quant Funds	Standard AI Trading Bots	Kadirov Quant Lab
Adaptability	Manual recalibration (Weeks)	Static rule retraining (Days)	Real-time self-correction (Minutes)
Workflow	Siloed teams & tools	Black-box execution	Unified Agentic Loop
Risk Guardrails	Rigid, hard-coded limits	Unpredictable in anomalies	Mathematical HITL Constraints

Business & Monetization Model

Kadirov Quant Lab operates on a dual-revenue architecture designed for high-margin scalability:

1. **Proprietary Trading / Fund Management:** Operating as a next-generation hedge fund, charging a standard management and performance fee structure (e.g., 2/20) on Assets Under Management (AUM).
2. **B2B Enterprise Licensing:** Offering the core hierarchical multi-agent framework as a SaaS/PaaS solution for institutional asset managers and family offices seeking to upgrade their legacy infrastructure.